VIVEK CHAURASIYA

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EDUCATION

Kanpur Institute of Technology (KIT) **2023 - 2027**
Bachelor of Technology in Computer Science & Engineering (AI & ML) *Kanpur, India*

- **Academic Record:** SGPA: **8.48/10** (Semester 6) | Cumulative CGPA: 7.0/10 across core CS & AI coursework.
- **Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Machine Learning.

TECHNICAL EXPERIENCE

Pradite **June 2026 - Present**
Co-Founder & Growth Lead *Kanpur, India*

- Co-founded zero-knowledge BYOK AI platform protecting developer code via client-side encryption (libsodium); drove initial developer onboarding and feedback loops.
- Built product distribution pipeline and community channels across student and indie developer clusters; synthesized bug reports to prioritize technical roadmap sprints.

India Space Academy **May 2025 - July 2025**
Remote Sensing & GIS Software Engineering Intern *Remote*

- Engineered client-side simulation engines for Vegetation Condition Index (VCI) drought severity mapping, Urban Heat Island analysis, and watershed prioritization using Digital Elevation Models (DEM).
- Translated satellite imagery processing algorithms into lightweight, dependency-free JavaScript modules packaged as single HTML files for offline use in low-bandwidth environments.

PROJECTS

Suraksha Bandhan: Real-Time Disaster Early Warning PWA | *JavaScript, Leaflet, PWA* **2026**

- Built an offline-first Progressive Web App (PWA) delivering real-time flood risk predictions and emergency alerts across 6 Indian languages with zero external backend dependencies.
- Integrated browser Geolocation API with Leaflet spatial mapping and OpenStreetMap Overpass API to calculate real-time river flood severity (High/Med/Low) based on precipitation and elevation data.
- Shortlisted in hackathon competition; implemented emergency SOS dispatch triggering pre-configured emergency services with zero latency.

Exam Seating Auto-Allocator | *Python, Flask, SQLite, Genetic Algorithms* **2025**

- Automated university examination hall allocation as a Constraint Satisfaction Problem (CSP) enforcing room capacity, branch mixing, and disability accommodation rules.
- Implemented genetic optimization algorithm with tailored crossover/mutation operators, cutting manual examination scheduling from 6 hours to under 15 minutes.
- Built a Flask administration portal generating official seat matrix PDFs; adopted by KIT examination cell in production across 2 consecutive semesters.

FunLangLab & HindiCode: Vernacular Transpiler & Web IDE | *JavaScript, AST Parser, Claude API* **2025**

- Developed a recursive descent parser and AST-based compiler supporting native Hindi and hybrid syntax, transpiling educational programming constructs directly into executable JavaScript.
- Engineered an in-browser execution sandbox with Claude API integration for automated syntax hint generation and natural-language error translation.
- Conducted user testing with 15 students at Barabanki secondary school; 80% of novice students wrote independent loops within 30 minutes.

Smart IoT Water Level & Telemetry Monitor | *C++, Arduino Uno, ESP8266, Blynk IoT* **2024 - 2025**

- Architected an automated water level telemetry system using HC-SR04 ultrasonic sensors calibrated to cylindrical geometry, transmitting metrics via ESP8266 Wi-Fi over MQTT.
- Deployed production units across 2 campus hostel overhead tanks; prevented 3 overflow incidents in month one and cut water wastage by ~20%.

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), C++, Java, C, SQL (MySQL, SQLite), HTML5/CSS3
Frameworks & Web:	Node.js, Express.js, Flask, React.js basics, RESTful APIs, Leaflet.js, Progressive Web Apps (PWA)
Embedded & Systems:	Arduino Uno/Nano, ESP8266/ESP32, Raspberry Pi, Sensor Interfacing (Ultrasonic, DHT, PIR), MQTT, CoAP
Tools & Databases:	Git, GitHub, Docker basics, Linux/Unix Shell, MongoDB, MySQL, SQLite, Firebase, Postman, VS Code

LEADERSHIP & RECOGNITION

- **Hackathon Shortlist (Suraksha Bandhan):** Shortlisted for real-time disaster early-warning system prototype.
- **Production Institutional Deployment:** Exam Seating Auto-Allocator adopted and deployed by KIT Examination Cell.
- **Technical Club Mentorship:** Guided 40+ junior engineering students in micro-controller setup, Arduino debugging, and IoT prototyping.